

Section B: Answer ALL questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided. Take $g = 9.81 \text{ m s}^{-2}$.

1. A bottle of water of temperature 4°C is taken out from a refrigerator and placed on a table. The room temperature is 30°C .



Figure 1.1

- (a) State one way of heat transfer from the surroundings to the bottle of water. (1 mark)

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- (b) Explain whether wrapping the bottle with an aluminum foil will speed up the rate of temperature increase. (2 marks)

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- (c) After a period of time, water droplets form on the surface of the bottle.

- (i) State the source of the water droplets. (1 mark)

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- (ii) Suppose 0.40 g of water droplets are collected on the surface of the bottle. Estimate the latent heat released from these droplets. (2 marks)

Given: specific latent heat of vaporization of water $= 2.26 \times 10^6 \text{ J kg}^{-1}$

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Answers written in the margins will not be marked.

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